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Temperature measurement at one or more cutting elements of a drill bit

Abstract

Methods and systems are disclosed that use a temperature sensor integral to a drill bit during drilling. The temperature sensor is configured to measure temperature associated with a cutting element of the drill bit over time during drilling. The temperature sensor is used to generate temperature data representing temperature of the cutting element of the drill bit over time during drilling. The temperature data is processed to determine and monitor at least one condition of the wellbore and/or the drill bit during drilling, such as i) detection of faults or openings or cracks or other geological rock features of the wellbore being drilled, ii) estimation of one or more drilling parameters of the drill bit (such as rate of penetration (ROP), iii) detection of lost cutting element(s), iv) detection of bit balling, and v) detection of drilling efficiency.

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Background/Summary

FIELD

(1) The present disclosure relates to while drilling systems and methods that detect and monitor conditions of a drill bit and/or a wellbore drilled by a drill bit.

BACKGROUND

(2) Cutting tools, such as downhole drill bits, are used to drill wellbores into the earth to access hydrocarbon reservoirs (e.g., oil and gas reservoirs) and non-hydrocarbon reservoirs (e.g., geothermal reservoirs). The drill bits have relatively long service lives with relatively infrequent failure, but are expensive to design and manufacture.

(3) While drilling a wellbore, the wellbore and/or the drill bit can experience a number of conditions that can impede the drilling operations and possibly necessitate expensive remedial operations. For example, lost circulation of drilling fluid can occur when the drilling fluid, known commonly as “mud”, flows into faults or openings or cracks or other geological rock features of the wellbore being drilled. Lost circulation is typically remediated by reducing the density of the drilling fluid, adding lost circulation material (LCM) to the drilling fluid, and/or pumping LCM

pills into the wellbore. These operations add both time and cost to the drilling operation. In another example, the rate of penetration (ROP) of the drill bit or other drilling parameter can deviate from the intended or target ROP value. Specifically, there are a large number of parameters that influence ROP, including rock strength of the formation, revolutions per minute (RPM), and weight on bit (WOB) and the flow rate and characteristics of drilling fluid. Some of these parameters are controllable and some cannot be controlled. In yet another example, one or more cutter elements integral to the drill bit can separate from the drill bit during drilling due to excessive vibration and dynamic loading experienced by the drill bit during drilling. This condition is typically referred to as a lost cutter and can be remediated by a fishing operation to retrieve the lost cutter. If the fishing operation fails, sidetrack drilling operations can be performed in order to drill around the portion of the wellbore which includes the lost cutter. In still another example, the drill bit can experience bit balling while drilling. Bit balling is an undesirable adherence and clinging of clay based drilled solids to the metal surfaces of the drill bit. It is usually caused by the hydration and/or dispersion of clay particles. The bit balling is typically remediated by increasing the weight on the bit, adding chemicals to the drilling fluid or perhaps stopping the drilling and pulling the drill bit of the wellbore to clean the bit. While drilling a wellbore, the wellbore and/or the drill bit can experience a number of conditions that can optimize the drilling operations. For example, drilling efficiency represents a condition where an optimal (high) penetration rate is achieved during drilling. Maintaining drilling efficiency can thus reduce the time and costs of the drilling operation. It can be difficult to detect and monitor these different conditions experienced by the wellbore and/or the drill bit during drilling with accuracy and confidence.

SUMMARY

(4) Methods and systems are disclosed that use a temperature sensor integral to a drill bit during drilling. The temperature sensor is configured to measure temperature associated with a cutting element of the drill bit over time during drilling. The temperature sensor is used to generate temperature data representing temperature of the cutting element of the drill bit over time during drilling. The temperature data is processed to determine and monitor at least one condition of the wellbore and/or the drill bit during drilling.

(5) In embodiments, the temperature sensor can include a thermocouple, a resistance temperature detector (RTD), thermistor, semiconductor-based temperature sensor, infrared temperature sensor, thermometer, bimetallic sensor, change-of-state sensor, silicon diode, or other type of temperature sensor.

(6) In embodiments, the temperature sensor can be configured to measure temperature representative of temperature at or near a cutting surface of the cutting element.

(7) In embodiments, the processing of the temperature data can be performed by a downhole system or a surface-located system.

(8) In embodiments, the at least one condition can relate to detection of one or more faults or openings or cracks or other geological rock features of the wellbore being drilled.

(9) In embodiments, the one or more faults or openings or cracks or other geological rock features of the wellbore being drilled can be detected by fluctuations in the temperature data over time.

(10) In embodiments, the at least one condition can relate to estimation of at least one drilling parameter (such as rate of penetration (ROP)) of the drill bit.

(11) In embodiments, the drilling parameter can be estimated by using a computational model of the drilling parameter as a function of measured temperature.

(12) In embodiments, the computational model can be calibrated for a given combination of rock type and cutting element or bit type.

(13) In embodiments, the at least one condition can relate to detection of at least one lost cutting element.

(14) In embodiments, the at least one lost cutting element can be detected by step changes in the temperature data over time.

- (15) In embodiments, the at least one condition can relate to detection of bit balling.
 - (16) In embodiments, the bit balling can be detected by a gradual variation in the temperature data over time.
 - (17) In embodiments, the at least one condition can relate to detection of drilling efficiency.
 - (18) In embodiments, the drilling efficiency can be detected by changes in the temperature data over time.
 - (19) This summary is provided to introduce a selection of concepts that are further described below in the detailed description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The subject disclosure is further described in the detailed description which follows, in reference to the noted plurality of drawings by way of non-limiting examples of the subject disclosure, in which like reference numerals represent similar parts throughout the several views of the drawings, and wherein:
- (2) FIG. 1 is a schematic diagram illustrating a drilling system for drilling an earth formation to form a wellbore, according to at least one embodiment of the present disclosure;
- (3) FIG. 2A is a bottom view of a drill bit, according to at least one embodiment of the present disclosure;
- (4) FIG. 2B is a partial cross-sectional view of the drill bit of FIG. 2A;
- (5) FIG. 3 is an image of part of a drill bit, according to at least one embodiment of the present disclosure;
- (6) FIG. 4 is a flowchart of a method for real-time detection and monitoring of condition(s) of a wellbore and/or drill bit during drilling based on temperature measurements made by at least one temperature sensor integral to the drill bit, according to at least one embodiment of the present disclosure;
- (7) FIG. 5A depicts plots of temperature data representing temperature of the gauge cutting element of the drill bit of FIG. 3 and RPM of the drill bit over time during a drilling experiment that follows the workflow of FIG. 4;
- (8) FIG. 5B depicts plots of temperature data representing temperature of the conical cutting element of the drill bit of FIG. 3 and RPM of the drill bit over time during the same drilling experiment of FIG. 5A;
- (9) FIG. 6A is a plot of temperature data representing temperature of the conical cutting element of the drill bit of FIG. 3 as a function of RPM of the drill bit during the drilling experiment of FIGS. 5A and 5B;
- (10) FIG. 6B depict plots of temperature data representing temperature of the gauge cutting element and conical cutting element, respectively, of the drill bit of FIG. 3 as a function of RPM of the drill bit during the drilling experiment of FIGS. 5A and 5B;
- (11) FIG. 7 is a plot of temperature data representing temperature of the conical cutting element of the drill bit of FIG. 3 as a function of ROP of the drill bit during the drilling experiment of FIGS. 5A and 5B;
- (12) FIG. 8 is an image of a drilling experiment performed on a Lazonby rock with four (4) small lateral holes disposed along the depth dimension of the rock. This drilling experiment aims to demonstrate that these holes can be detected with temperature measurements for one or more cutting elements of a drill bit as described herein;
- (13) FIG. 9 is a plot of temperature data representing temperature of the conical cutting element of

the drill bit of FIG. 3 and RPM of the drill bit over time during the drilling experiment of FIG. 8; (14) FIG. 10 is an image of a drilling experiment that is designed to demonstrate that one or more lost cutting elements can be detected with temperature measurements for one or more cutting elements of a drill bit;

(15) FIGS. 11A and 11B depict plots of temperature data obtained from a drilling experiment that involved drilling first and second wellbores into the same or similar Lazonby rock with common drilling parameters (e.g., RPM, ROP, WOB). The first wellbore is drilled in the rock with all cutting elements of the drill bit in place. The second wellbore is drilled in two parts. In this first part, the first half of the wellbore is drilled with all cutting elements of the drill bit in place. In the second part, two cutting elements are removed from the drill bit as shown in FIG. 10 and the second half of the wellbore is drilled. FIG. 11A depict plots of temperature data representing temperature of a gauge cutting element of the drill bit over time during the drilling experiment for the first and second wellbores. FIG. 11B depicts plots of temperature data representing temperature of a conical cutting element of the drill bit over time during the drilling experiment for the first and second wellbores; and

(16) FIG. 12 is a schematic block diagram of a computer processing system.

DETAILED DESCRIPTION

(17) The particulars shown herein are by way of example and for purposes of illustrative discussion of the embodiments of the subject disclosure only and are presented in the cause of providing what is believed to be the most useful and readily understood description of the principles and conceptual aspects of the subject disclosure. In this regard, no attempt is made to show structural details in more detail than is necessary for the fundamental understanding of the subject disclosure, the description taken with the drawings making apparent to those skilled in the art how the several forms of the subject disclosure may be embodied in practice. Furthermore, like reference numbers and designations in the various drawings indicate like elements.

(18) Embodiments of the present disclosure relate to real-time detection and monitoring of condition(s) of a wellbore and/or drill bit during drilling based on temperature measurements made by at least one temperature sensor integral to the drill bit. The temperature sensor may be configured to measure temperature at or near the cutting surface of a cutting element during drilling. The temperature measurements may be associated with time and/or depth of the drill bit. This may help a drilling operator dynamically adjust or control the drilling (for example, by adjusting one or more drilling parameters) based on the condition(s) of the wellbore and/or drill bit as determined from the real-time detection and monitoring.

(19) FIG. 1 shows a drilling system 100 for drilling an earth formation 101 to form a wellbore 102. The drilling system 100 includes a drill rig 103 used to turn a drilling tool assembly 104 which extends downward into the wellbore 102. The drilling tool assembly 104 may include a drill string 105, a bottomhole assembly (“BHA”) 106, and a drill bit 110, attached to the downhole end of drill string 105.

(20) The drill string 105 may include several joints of drill pipe 108 connected end-to-end through tool joints 109. The drill string 105 transmits drilling fluid through a central bore and transmits rotational power from the drill rig 103 to the BHA 106. In some embodiments, the drill string 105 may further include additional components such as subs, pup joints, etc. The drill pipe 108 provides a hydraulic passage through which drilling fluid is pumped from the surface. The drilling fluid discharges through selected-size nozzles, jets, or other orifices in the drill bit 110 for the purposes of cooling the bit 110 and cutting structures thereon, and for lifting cuttings out of the wellbore 102 as it is being drilled.

(21) The BHA 106 may include the drill bit 110 or other components. An example BHA 106 may include additional or other components (e.g., coupled between the drill string 105 and the drill bit 110). Examples of additional BHA components include drill collars, stabilizers, measurement-while-drilling (“MWD”) tools, logging-while-drilling (“LWD”) tools, downhole motors,

underreamers, section mills, hydraulic disconnects, jars, vibration or dampening tools, other components, or combinations of the foregoing. The BHA **106** may further include a rotary steerable system (RSS). The RSS may include directional drilling tools that change direction of the drill bit **110**, and thereby the trajectory of the wellbore. At least a portion of the RSS may maintain a geostationary position relative to an absolute reference frame, such as gravity, magnetic north, and/or true north. Using measurements obtained with the geostationary position, the RSS may locate the drill bit **110**, change the course of the bit **110**, and direct or steer the BHA **106** and drill bit **110** on a projected trajectory.

(22) In general, the drilling system **100** may include other drilling components and accessories, such as special valves (e.g., kelly cocks, blowout preventers, and safety valves). Additional components included in the drilling system **100** may be considered a part of the drilling tool assembly **104**, the drill string **105**, or a part of the BHA **106** depending on their locations in the drilling system **100**.

(23) The drill bit **110** in the BHA **106** may be any type of bit suitable for degrading down hole materials. For instance, the drill bit **110** may be a drill bit suitable for drilling the earth formation **101**. Example types of drill bits used for drilling earth formations are fixed-cutter or drag drill bits. In other embodiments, the drill bit **110** may be a mill used for removing metal, composite, elastomer, other materials downhole, or combinations thereof. For instance, the drill bit **110** may be used with a whipstock to mill into casing **107** lining the wellbore **102**. The drill bit **110** may also be a junk mill used to mill away tools, plugs, cement, other materials within the wellbore **102**, or combinations thereof. Swarf or other cuttings formed by use of a mill may be lifted to surface, or may be allowed to fall downhole.

(24) In accordance with at least one embodiment of the present disclosure, temperature data representing temperature measurements made by at least one temperature sensor integral to the drill bit during drilling can be obtained and communicated to a downhole system or surface-located system, and the temperature data can be processed to determine and monitor condition(s) of the wellbore and/or the drill bit during drilling, such as i) detection of faults or openings or cracks or other geological rock features of the wellbore being drilled, ii) estimation of drilling parameters of the drill bit (such as rate of penetration (ROP)), iii) detection of lost cutting element(s), iv) detection of bit balling, and v) detection of drilling efficiency.

(25) In one example, faults or openings or cracks or other geological rock features of the wellbore being drilled can be detected by fluctuations in the temperature data over time.

(26) In another example, one or more drilling parameters (such as rate of penetration (ROP)) can be estimated by using a computational model of the relevant drilling parameter as a function of measured temperature. The computational model can be calibrated for a given combination of rock type and cutting element type. In embodiments, the computational model can include a linear function or equation or a lookup table embodied by computer software. The computational model can be validated by experiments that show a linear change in measured temperature with respect to the drilling parameter for a given combination of rock type and cutting element type. During drilling, the rock type of the formation interval that is being drilled and the cutting element type or bit type for the cutting element associated with the temperature data can be identified and used to select a computational model corresponding to the identified combination of rock type and cutting element/bit type. The selected computation model can be used to determine the relevant drilling parameter (such as rate of penetration (ROP)) based on the temperature data of the cutting element as input.

(27) In yet another example, one or more lost cutting elements can be detected by step changes in the temperature data over time.

(28) In still another example, bit balling can be detected by the gradual variation in the temperature data over time.

(29) In another example, drilling efficiency can be detected by changes in the temperature data over

time.

(30) FIGS. 2A and 2B illustrate an example drill bit **210** according to at least one embodiment of the present disclosure. The drill bit **210** may include conical or other non-planar cutting elements **212** as well as planar or gauge cutting elements **230** (e.g., PDC shear cutters). The drill bit **210** includes a bit head **216** and a pin **218**. In the embodiment shown, the bit head **216** may be secured to the pin **218** with a bolted connection, such as with one or more mechanical fasteners **232**.

(31) FIG. 2B is a cross-sectional view of the bit **210**. Cutting element **212** is instrumented with a temperature sensor **214** integral to the cutting element **212**. The cutting element **212** may have conical or ridged cutting surface as shown. Alternatively, the cutting element **212** can have a convex cutting surface, a concave cutting surface, a planar cutting surface, any other shaped cutting surface, and combinations thereof. The temperature sensor **214** may be configured to measure a temperature representative of the temperature at or near the cutting surface of the cutting element **212**. In the embodiment shown, the temperature sensor **214** is inserted into a body **213** of the cutting element **212**. In some embodiments, the body **213** may define a bore that extends into the body **213**. The bore may be located in a substrate of the cutting element **212**, such as a tungsten carbide substrate or other substrate. In some embodiments, the bore may extend into the insert of the cutting element **212**. For example, the bore may extend into a polycrystalline diamond (PCD) insert connected to the substrate. In embodiments, the bore can be formed by creating a small hole in the back of cutting element **212** and drilling through the hole into the superhard table of the cutting element **212** (e.g., into a polycrystalline diamond table). The temperature sensor **214** can then be inserted in this hole with the hole then filled with thermally conductive potting or paste such that the temperature sensor **214** is in good thermal contact with the diamond or other superhard table of the cutting element **212**.

(32) The temperature sensor **214** may be any type of temperature sensor. For example, the temperature sensor **214** may include a thermocouple, a resistance temperature detector (RTD), thermistor, semiconductor-based temperature sensor, infrared temperature sensor, thermometer, bimetallic sensors, change-of-state sensors, silicon diodes, any other types of temperature sensor, and combinations thereof.

(33) The temperature sensor **214** may be inserted into the bore. Inserting the temperature sensor **214** into the bore may allow the temperature sensor **214** to measure a temperature representative of the temperature at or near the cutting surface **215** of the cutting element **212**. Note that frictional and contact forces of the cutting surface **215** with the rock of the wellbore during drilling may increase the temperature of the cutting surface **215**.

(34) In other embodiments, the temperature sensor **214** may be disposed at another location relative to the cutting element **212**. For example, the temperature sensor **214** may be in contact with the cutting element **212**, or the temperature sensor **214** may be located adjacent to the cutting element **212**, or the temperature sensor **214** may be located close to the outer surface of the bit head **216** parallel to the cutting element **212**.

(35) The drill bit **210** includes a wiring conduit **220** that extends through at least a portion of the bit head **216** to the pin **218**. The wiring conduit **220** may allow wiring from the temperature sensor **214** to extend through the drill bit **210** to an electronics module **222** in the pin **218**. In some embodiments, the wiring conduit **220** may include a pipe or a sleeve that is inserted into the bit head **216** during manufacturing. In some embodiments, the bit head **216** may be additively manufactured, and the wiring conduit **220** and/or a void for the wiring conduit **220** may be formed during the additive manufacturing process. In some embodiments, the wiring conduit **220** may extend through an integrally formed portion of the bit head **216**. While embodiments of the present disclosure are directed to a bit pin **218**, it should be understood that the bit head **216** may be connected to any other connector, including a bit box.

(36) During assembly or manufacture of the bit head **216** and the pin **218**, the wiring conduit **220** may align with a pin conduit **224**. The wiring from the temperature sensor **214** may pass from the

wiring conduit **220** in the bit head **216** to the pin conduit **224**. The pin conduit **224** may allow the wiring to pass to the electronics module **222**. In this manner, the temperature sensor **214** may be connected to the electronics module **222** in the pin **218**. In some embodiments, the wiring conduit **220** may be aligned with the pin conduit **224** when the bit head **216** is secured to the pin **218**.

(37) In embodiments, the bit head **216** can be coupled to the pin **218** in any manner. For example, the bit head **216** may be connected to the pin **218** with a bolted connection. In some examples, the bit head **216** may be connected to the pin **218** with a threaded connection. In some examples, when the bit head **216** is coupled to the pin **218**, the wiring conduit **220** and the pin conduit **224** may be aligned. In some embodiments, a bolted connection between the bit head **216** and the pin **218** may facilitate the alignment of the wiring conduit **220** and the pin conduit **224**.

(38) As discussed herein, the drill bit **210** may include an electronics module **222** with one or more wires or cables connecting the temperature sensor **214** to the electronics module **222**. The wire(s) or cable(s) carry the electric signals that represent the temperature measurements performed by the temperature sensor **214**. In some embodiments, the electronics module **222** can collect and monitor data representing the temperature measurements made by the temperature sensor **214**. In some embodiments, the electronics module **222** may record or store data representing the temperature measurements in electronic memory.

(39) In the embodiment shown, the pin **218** contains one or more instrumentation pockets **226** or other features that can house or connect to electronics module **222**. The wiring conduit **220** may extend from the cutting element **212** to the instrumentation pocket **226**. In some embodiments, the electronics module **222** may be located in the instrumentation pocket **226**. In some embodiments, the instrumentation pocket **226** may include other sensors. For example, the one or more instrumentation pockets **226** may include a force sensor, a torque sensor, an accelerometer, a gyroscopic sensor, a pressure sensor, a temperature sensor to measure temperature of the drilling fluid, any other sensors, and combinations thereof. In some embodiments, the information from the one or more sensors in the instrumentation pocket **226** may be used to detect and monitor condition(s) of the wellbore and/or the drill bit as described herein.

(40) In embodiments, the drill bit **210** can include a fluid and/or pressure seal between the electronic components and the drilling fluid. For example, the bit head **216** may include a head seal **228** between the bit head **216** and the pin **218** at the interface between the wiring conduit **220** and the pin conduit **224**. The head seal **228** may include a sealing member, such as an O-ring or other sealing member. The head seal **228** may further include a resilient element. When the bit head **216** is secured to the pin **218**, the compressive force of the connection may compress the resilient element against the sealing member. The compressed sealing member may result in a seal between the bit head **216** and the pin **218**. This may help to reduce and/or prevent the ingress of drilling fluid and/or other contaminants into the wiring conduit **220** and the pin conduit **224**. In some embodiments, the resilient element may be a frustoconical resilient elements, such as a Belleville washer. In some embodiments, the resilient element may include a wave spring or other resilient element. In some embodiments, the instrumentation pocket **226** may include a seal between the instrumentation pocket **226** and the borehole annulus to reduce or prevent the ingress of drilling fluid and/or other contaminants into the instrumentation pocket **226**.

(41) In the embodiment shown, the bit drill **210** includes multiple cutting elements, including cutting element **212** and cutting element **230**. One or more of the cutting elements may be PDC cutters or have other compositions. The drill bit **210** can be equipped with additional temperature sensor(s) that are similar to temperature sensor **214** as described herein. Such additional temperature sensor(s) can be configured to measure temperature at the cutting surface of other cutting elements, such as cutting element **230**. The bit head **216** and the pin **218** can be configured with additional conduits that carry one or more wires or cables connecting the additional temperature sensor(s) to the electronics module **222**.

(42) In embodiments, the electronics module **222** can communicate with a downhole device. For

example, the electronics module **222** can communicate with a MWD module, such as through a wired and/or a wireless connection. The MWD module may receive data representing the temperature measurements performed by the temperature sensor(s) integral to the drill bit **210**.

(43) In embodiments, the MWD module or other downhole system (which can include module **222**) can be configured to collect and process data representing the temperature measurements in real-time during drilling to determine and monitor conditions of the wellbore being drilling by the drill bit **210** during the drilling, such as i) detection of faults openings or cracks, ii) estimation of rate of penetration, iii) detection of lost cutting element(s), iv) detection of bit balling, and v) detection of drilling efficiency. The MWD module or other downhole system can then communicate data representing such condition(s) to a surface-located system for output to a drilling operator. For example, the MWD module may communicate the data representing such condition(s) using any downhole communication system, including mud pulse telemetry, electromagnetic communication, wired drill pipe, drill pipe telemetry, any other downhole communication system, and combinations thereof. The surface-located system can output or otherwise convey the condition(s) of the wellbore and/or drill bit **210** as determined from the monitoring via visual and/or audible alerts presented to the drilling operator. In embodiments, the drilling operator can dynamically adjust or control the drilling (for example, by adjusting one or more drilling parameters) based on the condition(s) of the wellbore and/or the drill bit **210** as represented by the data representing such condition(s).

(44) In other embodiments, the MWD module or other downhole system (which can include module **222**) can be configured to collect data representing the temperature measurements and communicate such data to a surface-located system. For example, the MWD module may communicate the data representing the temperature measurements using any downhole communication system, including mud pulse telemetry, electromagnetic communication, wired drill pipe, drill pipe telemetry, any other downhole communication system, and combinations thereof. The surface-located system can be configured to process the data representing the temperature measurements to determine and monitor condition(s) of the wellbore being drilled by the drill bit **210** during drilling, such as i) detection of faults openings or cracks, ii) estimation of rate of penetration, iii) detection of lost cutting element(s), iv) detection of bit balling, and v) detection of drilling efficiency. The surface-located system can output or otherwise convey the condition(s) of the wellbore and/or the drill bit **210** as determined from the monitoring via visual and/or audible alerts presented to the drilling operator. In embodiments, the drilling operator can dynamically adjust or control the drilling (for example, by adjusting one or more drilling parameters) based on the condition(s) of the wellbore and/or the drill bit **210** as determined from the monitoring.

(45) FIG. **3** is an image of part of a fixed-cutter drill bit **310** that has a blade with a gauge cutting element instrumented with a temperature sensor as well as a cone cutting element instrumented with a temperature sensor in a manner similar to that described above with respect to FIGS. **2A** and **2B**.

(46) FIG. **4** is a flowchart that implements a method for real-time detection and monitoring of condition(s) of a wellbore and/or drill bit during drilling based on temperature measurements made by at least one temperature sensor integral to the drill bit.

(47) The operations begin in block **401** by configuring at least one temperature sensor integral to a drill bit to measure temperature at or near the cutting surface of one or more cutting elements of the drill bit during drilling.

(48) In block **403**, data representing the measured temperature of **401** is collected/stored in electronic memory.

(49) In block **405**, the temperature data of **403** is communicated to a downhole system or surface-located system.

(50) In block **407**, the downhole system or surface-located system of **405** is configured to process the temperature data to monitor conditions of the wellbore and/or the drill bit during drilling, such

as i) detection of faults openings or cracks, ii) estimation of one or more drilling parameters (such as ROP), iii) detection of lost cutting element(s), iv) detection of bit balling, and v) detection of drilling efficiency.

(51) For example, as part of block **407**, faults or openings or cracks or other geological rock features of the wellbore being drilled can be detected by fluctuations in the temperature data representing temperature of one or more cutting elements over time.

(52) In another example, as part of block **407**, one or more drilling parameters (such as rate of penetration (ROP)) can be estimated by using a computational model of the drilling parameter as a function of measured temperature. The computational model can be calibrated for a given combination of rock type and cutting element type. In embodiments, the rock type of the formation interval that is being drilled and the cutting element type for the cutting element associated, the temperature data can be identified and used to select a computational model corresponding to the identified combination of rock type and cutting element type. The selected computation model can be used to determine the relevant drilling parameter (such as rate of penetration (ROP)) based on the temperature data representing temperature of the cutting element as input.

(53) In yet another example, as part of block **407**, one or more lost cutting elements can be detected by step changes in the temperature data representing temperature of one or more cutting elements over time. The lost cutting element(s) can be separated from the drill bit during drilling due to excessive vibration and dynamic loading experienced by the drill bit during drilling.

(54) In still another example, as part of block **407**, bit balling can be detected by a gradual increase or variation in the temperature data representing temperature of one or more cutting elements over time. Bit balling is an undesirable adherence and clinging of clay based drilled solids to the metal surfaces of the drill bit. It is usually caused by the hydration and/or dispersion of clay particles.

(55) In another example, as part of block **407**, drilling efficiency can be detected by changes in the temperature data representing temperature of one or more cutting elements over time. Drilling efficiency represents a condition where an optimal (high) penetration rate is achieved during drilling.

(56) FIG. 5A depicts plots of temperature data representing temperature of the gauge cutting element of the drill bit **301** of FIG. 3 and RPM of the drill bit **301** over time during a drilling experiment that follows the workflow of FIG. 4. FIG. 5B depicts plots of temperature data representing temperature of the conical cutting element of the drill bit **301** of FIG. 3 and RPM of the drill bit **301** over time during the same drilling experiment of FIG. 5A. In the drilling experiment of FIGS. 5A and 5B, the RPM of the drill bit **301** can be determined by one or more downhole sensors, such as a gyroscope, which is part of the BHA of the drilling system as is well known.

(57) FIG. 6A is a plot of temperature data representing temperature of the conical cutting element of the drill bit **301** of FIG. 3 as a function of RPM of the drill bit **301** during the drilling experiment of FIGS. 5A and 5B. FIG. 6B depict plots of temperature data representing temperature of the gauge cutting element and conical cutting element, respectively, of the drill bit **301** of FIG. 3 as a function of RPM of the drill bit **301** during the drilling experiment of FIGS. 5A and 5B. The top plot represents the temperature measurements for the gauge cutting element as a function of RPM of the drill bit **301**. The bottom plot represents the temperature measurements for the conical cutting element as a function of RPM of the drill bit **301**. Note that the temperatures of the cutting elements both depend linearly on RPM. However the slope of the curve and the offset at y axis are most likely dependent on rock type, and cutter type/size. Thus, a linear function can be fitted to the experimental temperature data of the top plot to derive a computational model that relates the temperature measurement for the gauge type cutting element to rate of penetration (ROP) of the drill bit during drilling. Similarly, a linear function can be fitted to the experimental temperature data of the bottom plot to derive a computational model that relates the temperature measurement for the conical type cutting element to rate of penetration (ROP) of the drill bit during drilling. The

computation model(s) can be specific to a given rock type. The drilling experiments can be repeated for different rock types to derive computational models that relate cutting element temperature to drill bit ROP for different combinations of cutting element type and rock type. During drilling, the rock type of the formation interval that is being drilled and the cutting element type (such as conical type or gauge type for a given bit) for the cutting element associated with the temperature data can be identified and used to select the computational model corresponding to the identified combination of rock type and cutting element type. The selected computation model can be used to determine the rate of penetration (ROP) of the drill bit based on the temperature data of the cutting element as input.

(58) FIG. 7 is a plot of temperature data representing temperature of the conical cutting element of the drill bit **301** of FIG. 3 as a function of ROP of the drill bit **301** during the drilling experiment of FIGS. 5A and 5B. Note that the measured temperature of the conical cutting element depends linearly on ROP. Thus, a linear function can be fitted to the experimental temperature data to derive a computational model that relates the temperature measurement for the conical type cutting element to ROP of the drill bit during drilling. Similarly, a linear function can be fitted to the experimental temperature data to derive a computational model that relates the temperature measurement for the gauge type cutting element to ROP of the drill bit during drilling. The computation model(s) can be specific to a given rock type. The drilling experiments can be repeated for different rock types and different bits or bit types to derive computational models that relate cutting element temperature to drill bit ROP for different combinations of cutting element type/bit type and rock type.

(59) In embodiments, the computational model for a particular combination of bit type (cutter type/size) and rock type can be derived from experiments performed at various ROP values that measure corresponding cutter temperatures. Once a plot similar to FIG. 7 is obtained, the slope and offset of the curve can be calculated. This curve can be then used to predict rate of penetration (ROP) of the drill bit while drilling in a real drilling operation, given that the same or similar drill bit (with the same or similar cutter as used in the experiments is instrumented with a temperature sensor) is drilling the same or similar rock type. As temperature data is obtained during drilling, corresponding ROP can be predicted from the curve.

(60) FIG. 8 illustrates a drilling experiment performed on a Lazonby rock with four (4) small lateral holes disposed along the depth dimension of the rock. This drilling experiment aims to demonstrate that these holes can be detected with temperature measurements for one or more cutting elements of a drill bit as described herein.

(61) FIG. 9 is a plot of temperature data representing temperature of the conical cutting element of the drill bit **301** of FIG. 3 and RPM of the drill bit **301** over time during this drilling experiment. This plot shows that the four (4) lateral holes can be detected by processing the temperature data to identify fluctuations in the temperature data over time that are representative of the geological rock features. Similar processing can be used to identify faults or openings or cracks or other geological rock features of the wellbore being drilled.

(62) Another drilling experiment was performed that involved drilling first and second wellbores into the same or similar Lazonby rock with common drilling parameters (e.g., RPM, ROP, WOB). The first wellbore is drilled in the rock with all cutting elements of the drill bit in place. The second wellbore is drilling in two parts. In this first part, the first half of the wellbore is drilled with all cutting elements of the drill bit in place. In the second part, two cutting elements are removed from the drill bit as shown in FIG. 10 and the second half of the wellbore is drilled. FIG. 11A depicts plots of temperature data representing temperature of a gauge cutting element of the drill bit over time during the drilling experiment for the first and second wellbores. The top plot shows the temperature measurements for the gauge cutting element of the drill bit over time while drilling the first wellbore. The bottom plot shows the temperature measurements for the same gauge cutting element of the drill bit over time while drilling the second wellbore. FIG. 11B depicts plots of

temperature data representing temperature of a conical cutting element of the drill bit over time during the drilling experiment for the first and second wellbores. The top plot shows the temperature measurements for a conical cutting element of the drill bit over time while drilling the first wellbore. The bottom plot shows the temperature measurements for the same conical cutting element of the drill bit over time while drilling the second wellbore. The plots of FIGS. **11A** and **11B** show that lost cutting element(s) can be detected by processing the temperature data to identify increased temperature in the temperature data for the respective cutting element(s) which is representative of the presence of one or more lost cutting element(s).

(63) FIG. **12** illustrates an example device **2500**, with a processor **2502** and memory **2504** that can be configured to implement part(s) of methods and workflows as discussed in the present disclosure. Memory **2504** can also host one or more databases and can include one or more forms of volatile data storage media such as random-access memory (RAM), and/or one or more forms of nonvolatile storage media (such as read-only memory (ROM), flash memory, and so forth).

(64) Device **2500** is one example of a computing device or programmable device and is not intended to suggest any limitation as to scope of use or functionality of device **2500** and/or its possible architectures. For example, device **2500** can comprise one or more computing devices, programmable logic controllers (PLCs), etc.

(65) Further, device **2500** should not be interpreted as having any dependency relating to one or a combination of components illustrated in device **2500**. For example, device **2500** may include one or more of: computers, such as a laptop computer, a desktop computer, a mainframe computer, etc., or any combination or accumulation thereof.

(66) Device **2500** can also include a bus **2508** configured to allow various components and devices, such as processors **2502**, memory **2504**, and local data storage **2510**, among other components, to communicate with each other.

(67) Bus **2508** can include one or more of any of several types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures. Bus **2508** can also include wired and/or wireless buses.

(68) Local data storage **2510** can include fixed media (e.g., RAM, ROM, a fixed hard drive, etc.) as well as removable media (e.g., a flash memory drive, a removable hard drive, optical disks, magnetic disks, and so forth). One or more input/output (I/O) device(s) **2512** may also communicate via a user interface (UI) controller **2514**, which may connect with I/O device(s) **2512** either directly or through bus **2508**.

(69) In one possible implementation, a network interface **2516** may communicate outside of device **2500** via a connected network. A media drive/interface **2518** can accept removable tangible media **2520**, such as flash drives, optical disks, removable hard drives, software products, etc. In one possible implementation, logic, computing instructions, and/or software programs comprising elements of module **2506** may reside on removable media **2520** readable by media drive/interface **2518**.

(70) In one possible embodiment, input/output device(s) **2512** can allow a user (such as a human annotator) to enter commands and information to device **2500**, and also allow information to be presented to the user and/or other components or devices. Examples of input device(s) **2512** include, for example, sensors, a keyboard, a cursor control device (e.g., a mouse), a microphone, a scanner, and any other input devices known in the art. Examples of output devices include a display device (e.g., a monitor or projector), speakers, a printer, a network card, and so on.

(71) Various systems and processes of present disclosure may be described herein in the general context of software or program modules, or the techniques and modules may be implemented in pure computing hardware. Software generally includes routines, programs, objects, components, data structures, and so forth that perform particular tasks or implement particular abstract data types. An implementation of these modules and techniques may be stored on or transmitted across

some form of tangible computer-readable media. Computer-readable media can be any available data storage medium or media that is tangible and can be accessed by a computing device. Computer readable media may thus comprise computer storage media. “Computer storage media” designates tangible media, and includes volatile and non-volatile, removable, and non-removable tangible media implemented for storage of information such as computer readable instructions, data structures, program modules, or other data. Computer storage media include, but are not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other tangible medium which can be used to store the desired information, and which can be accessed by a computer.

(72) Some of the methods and processes described above can be performed by a processor. The term “processor” should not be construed to limit the embodiments disclosed herein to any particular device type or system. The processor may include a computer system. The computer system may also include a computer processor (e.g., a microprocessor, microcontroller, digital signal processor, general-purpose computer, special-purpose machine, virtual machine, software container, or appliance) for executing any of the methods and processes described above.

(73) The computer system may further include a memory such as a semiconductor memory device (e.g., a RAM, ROM, PROM, EEPROM, or Flash-Programmable RAM), a magnetic memory device (e.g., a diskette or fixed disk), an optical memory device (e.g., a CD-ROM), a PC card (e.g., PCMCIA card), or other memory device.

(74) Some of the methods and processes described above can be implemented as computer program logic for use with the computer processor. The computer program logic may be embodied in various forms, including a source code form or a computer executable form. Source code may include a series of computer program instructions in a variety of programming languages (e.g., an object code, an assembly language, or a high-level language such as C, C++, or JAVA). Such computer instructions can be stored in a non-transitory computer readable medium (e.g., memory) and executed by the computer processor. The computer instructions may be distributed in any form as a removable storage medium with accompanying printed or electronic documentation (e.g., shrink wrapped software), preloaded with a computer system (e.g., on system ROM or fixed disk), or distributed from a server or electronic bulletin board over a communication system (e.g., the Internet or World Wide Web).

(75) Alternatively or additionally, the processor may include discrete electronic components coupled to a printed circuit board, integrated circuitry (e.g., Application Specific Integrated Circuits (ASIC)), and/or programmable logic devices (e.g., a Field Programmable Gate Arrays (FPGA)). Any of the methods and processes described above can be implemented using such logic devices.

(76) Although only a few example embodiments have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the example embodiments without materially departing from this invention.

(77) Accordingly, all such modifications are intended to be included within the scope of this disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures. Thus, although a nail and a screw may not be structural equivalents in that a nail employs a cylindrical surface to secure wooden parts together, whereas a screw employs a helical surface, in the environment of fastening wooden parts, a nail and a screw may be equivalent structures. It is the express intention of the applicant not to invoke 35 U.S.C. § 112, paragraph 6 for any limitations of any of the claims herein, except for those in which the claim expressly uses the words ‘means for’ together with an associated function.

Claims

1. A method for detecting and monitoring at least one condition of a wellbore and/or drill bit during drilling, the method comprising: drilling the wellbore with the drill bit, wherein the drill bit is instrumented with a temperature sensor configured to measure temperature associated with a cutting element of the drill bit over time during drilling; using the temperature sensor to generate temperature data representing temperature of the cutting element of the drill bit over time during drilling; and processing the temperature data to determine rate of penetration (ROP) of the drill bit during drilling, wherein the ROP of the drill bit is estimated using a computational model of drill bit ROP as a function of measured temperature, wherein the computational model is derived by fitting a linear function that relates drill bit ROP to temperature data, and wherein the computational model is calibrated for a given combination of rock type and cutting element or bit type.
 2. A method according to claim 1, wherein: the temperature sensor comprises a thermocouple, a resistance temperature detector (RTD), thermistor, semiconductor-based temperature sensor, infrared temperature sensor, thermometer, bimetallic sensor, change-of-state sensor, silicon diode, or other type of temperature sensor.
 3. A method according to claim 1, wherein: the temperature sensor is configured to measure temperature representative of temperature at or near a cutting surface of the cutting element.
 4. A method according to claim 1, wherein: the processing of the temperature data is performed by a downhole system or a surface-located system.
 5. A drilling system comprising: a drill bit for drilling wellbore through a formation, wherein the drill bit is instrumented with a temperature sensor configured to measure temperature associated with a cutting element of the drill bit over time during drilling; a processor configured to: generate or obtain temperature data representing temperature of the cutting element of the drill bit over time during drilling as measured by the temperature sensor; and process the temperature data to determine rate of penetration (ROP) of the drill bit during drilling, wherein the ROP of the drill bit is estimated using a computational model of drill bit ROP as a function of measured temperature, wherein the computational model is derived by fitting a linear function that relates drill bit ROP to temperature data, and wherein the computational model is calibrated for a given combination of rock type and cutting element or bit type.
 6. A drilling system according to claim 5, wherein: the temperature sensor comprises a thermocouple, a resistance temperature detector (RTD), thermistor, semiconductor-based temperature sensor, infrared temperature sensor, thermometer, bimetallic sensor, change-of-state sensor, silicon diode, or other type of temperature sensor.
 7. A drilling system according to claim 5, wherein: the temperature sensor is configured to measure temperature representative of temperature at or near a cutting surface of the cutting element.
 8. A drilling system according to claim 5, wherein: the processor comprises a downhole processor or a surface-located processor.
 9. A method for detecting and monitoring at least one condition of a wellbore during drilling, the method comprising: drilling the wellbore with the drill bit, wherein the drill bit is instrumented with a temperature sensor configured to measure temperature associated with a cutting element of the drill bit over time during drilling; using the temperature sensor to generate temperature data representing temperature of the cutting element of the drill bit over time during drilling; and processing the temperature data to detect one or more faults or openings or cracks in the wellbore by detecting fluctuations in the temperature data over time.
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